



Full length article

Multi-factor evaluation of clustering methods for e-commerce application

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ABSTRACT

This research aimed to investigate the application of Vlse Kriterijumska Optimizacija I Kompromisno Resenje (VIKOR) and Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS) multi-criteria decision-making to select the optimal clustering for e-commerce customer segmentation. In this context, clustering as an unsupervised machine learning method offered a way to overcome the limitations of traditional grouping, particularly by providing the ability to capture the diverse needs of consumers. A total of five different clustering methods were considered based on the behavioral data of e-commerce customers. Even though the analyzed algorithms were well-known and widely used, the comprehensive and multidirectional comparison was not trivial. Selected approaches were evaluated on the basis of twelve indicators (decision criteria), divided into four characteristics that take into account both the out-of-context aspects of clustering and the requirements arising from the context of using the clustering results. The results showed consistent outcomes from both analyzed Multi-Criteria Decision Methods, with some notable differences. The methods obtained the same ranking of the top three clustering algorithms (K-median - BIRCH - K-means). However, the TOPSIS and VIKOR sensitivity analysis recommended K-means in 87% of the cases and 60% of the variants verified, respectively. The parameterization of the decision factors had a significant impact on the final ranking of clustering options. This research demonstrated the practical application of the decision methods in selecting the best clustering for multivariate user interfaces to improve personalization in e-commerce.

1. Introduction

In the dynamic landscape of e-commerce, personalization is becoming critical in improving the user experience [1] and increasing conversion rates [2]. However, given the sheer number of unique customers, it is worth considering grouping them and targeting personalized communication to specific segments of users with similar characteristics rather than to individuals.

In e-commerce development, segmentation was originally based on decision rules. However, the current approaches often use machine learning (ML), which is a branch of artificial intelligence (AI), to improve the performance of data-based techniques. ML includes various clustering methods in e-commerce customer segmentation [3]. The primary advantage is the ability to handle large data sets as well as identify dependencies and relationships unobservable through decision-rule reasoning. In contrast, the main limitation is computational complexity, requiring high-performance servers.

Different clustering methods are used for e-commerce customer segmentation. The literature review reported that the most commonly

used algorithms are based on the K – approach, including K-means, K-medians, and K-medoids [4]. The most popular application use machine learning to generate product recommendations in e-commerce [5]. K-means is used in e-commerce for both customer [6,7] and product segmentation [8]. Other methods used to cluster e-commerce data such as behavioral, demographic and combined [9] include K-medians [10], K-medoids [11], K-mode [12], Fuzzy C-Means [13,14], Balanced Iterative Reducing and Clustering using Hierarchies (BIRCH) [15], Self-organizing Map (SOM) [16], Spectral Clustering [17], etc.

Due to the differences in clustering algorithms, the correct choice of method and parameterization is an important issue. However, the decision is not obvious since many factors could be considered. The most commonly used clustering quality criteria are Silhouette Score (SS), Davies-Bouldin Index (DBI), Calinski-Harabasz Index (CHI), and Dunn Index (DI). These criteria are used to compare different clustering methods for e-commerce applications. In the literature, comparisons using SS [18,19], DBI [20], and CHI [21] as decision factors are commonly reported. It is worth noting that, with the exception of the last study, which combined SS and CHI indicators, the other studies based

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the comparison of clustering methods (and the selection of the best one) on only one measure. In some cases, the decision model was more extensive and included various factors: speed, clustering accuracy, DBI, within/total (W/T) clustering error index, and statistical significance test [22], but the number of criteria considered was limited.

Another problem is selecting the optimal number of clusters for a fixed algorithm. CHI was also used [11] for this purpose, and the issue of parameterization sometimes includes the number of resulting clusters [23,24].

Factors related to computational complexity, which affect processing time and resource consumption, are analyzed less frequently. This feature appears in some research as the limitations or challenges [25]. The aspect is analyzed in combination with other characteristics for a more accurate comparison of available options [26]. Meanwhile, the use of server resources due to computational complexity is among the biggest differentiators of clustering algorithms [27].

The context of the use of clustering results is rarely considered but can be extremely important. The business purpose of clustering should not be overlooked when evaluating available algorithms. In the case of e-commerce customer segmentation, an example is the desire to obtain segments of the closest possible size, or a certain minimum number of users. This expectation is related to cost rationality because personalized communications may not be prepared for small segments. This limitation was crucial during the development of the platform for serving dedicated variants of the e-commerce user interface (UI) [28].

Multiple Criteria Decision Analysis (MCDA) can be used to select the optimal clustering option and consider groups of different decision factors. However, there is a lack of publications confirming the direct application of the method to the analysis of e-commerce behavioral data. MCDA is commonly used to compare groupings, such as in-retailer segmentation [29], and financial risk analysis [30]. Such methods are used to optimize distributed networks [31], cloud services [32], and renewable energy solutions [33]. Therefore, the range of application is broad, and the potential use for selecting the optimal clustering method has significant practical implications. An analysis of the literature leads to the conclusion that MCDA methods are used to compare clustering algorithms and offer the flexibility to consider various decision factors [34]. In addition, MCDA can also be applied to other AI/ML application areas, specifically when multiple criteria are considered or ranked. For example, this approach can potentially be adopted to optimize tools used to predict prices of different types of goods, namely natural resources [35], electrical energy [36], real estates [37], food [38], as well as the impact of sustainability issues [39].

This conclusion is used to identify the research gap. Among the various studies comparing clustering algorithms, few consider more than one decision criterion. In addition, there is a lack of works describing the application of formal methods of multi-criteria analysis in the process of selecting the optimal way to cluster e-commerce customers based on their detailed behavioral data, including not just the basic Recency, Frequency, and Monetary (RFM) model, but, every single user activity in an online store. The described research was designed since the verification of MCDA for the selection of methods used for clustering the behavioral data of e-commerce customers has not been reported before. The previous research [40] confirmed the effectiveness of simple MCDMs for selecting e-commerce customer clustering algorithms and identified directions for further research. The proposed improved and extended approach includes more advanced multi-criteria analysis techniques, adds more clustering algorithms, and expands the set of decision criteria. Based on behavioral data, clustering is performed using five algorithms and different numbers of clusters. VIKOR [41] and TOPSIS [42] methods were used to identify the optimal alternative for customer segmentation in e-commerce applications. Therefore, this research aims to provide insights into the strengths and weaknesses of different clustering methods, helping practitioners make informed decisions. Experimental research allowed the verification of

the selected algorithms, identified differences in the results obtained by MCDA, and enabled sensitivity analysis due to the decision factors.

The paper's contribution is threefold. First, it presents a framework including metrics corresponding to decision factors that address various aspects of clustering quality. These criteria contain traditional and commonly used measures, as well as indicators related to the business context of cluster use. This comprehensive approach to the distinguishing features of clustering methods is a practical contribution of the research. In addition, customer segmentation is based on very detailed behavioral data that goes beyond standard analysis based on orders placed or reviews left. This aspect is also novel, as more general data (e.g., on purchases made or products viewed) is usually used for this purpose. Second, it presents a comprehensive approach to multi-criteria clustering algorithm selection using TOPSIS and VIKOR methods to meet specific business requirements in e-commerce. This constitutes a theoretical contribution, as it shows the applicability of MCDA in the process of selecting a clustering approach, and a practical one, as it indicates the usefulness of the methods analyzed. Third, the paper includes a sensitivity analysis showing that considering only one aspect of clustering quality can lead to decisions that are different from those that should be made when many decision factors are considered.

2. Material and methods

The basis of the research discussed is a dataset of online clothing store customer behavior, including only unique user sessions. The structure contains information about identifiers (session, client, device, UI variant served), actions taken (the page, type, subtype, and attributes), and activity time (to determine the order of actions and the time spent on specific pages).

The information collected was based on clickstream data [45] which increases the suitability for segmenting e-commerce customers [46]. Therefore, the analysis uses information on detailed actions performed by the customer in the online store. It allowed for a better understanding of users compared to typical segmentations based on the RFM model. The main characteristics were the types of activities, the context, frequency, and effects. Fig. 1 shows an example of raw behavioral data in the form of a detailed clickstream.

The *AIM²* platform, developed by Fast White Cat S.A. of Poland, a vendor of e-commerce systems based on Adobe Magento, was used to collect behavioral data. The architecture of the solution is based on the Service Oriented Architecture (SOA) [47] paradigm to ensure maximum flexibility and openness.

The behavioral data was preprocessed with the goal of enriching it with product information, cleaning it, and preparing it for the use of clustering algorithms. This time-consuming and resource-intensive phase was crucial for the correct implementation of the study. However, since this was a common operation for all clustering methods analyzed, it was not included in the verification of differences in resource consumption. Therefore, it can be assumed that the final training data were prepared in the same way for all algorithms and thus provided identical conditions for the study.

The research conducted included five clustering methods, namely *Agglomerative*, *K – means*, *BIRCH*, *Genetic* and *K – medians*. These methods differ in clustering approach, parameterization, and algorithms, as presented in Table 1.

Since the purpose of the research was to verify methods of multi-criteria analysis and not to propose new clustering techniques, the focus was on classical algorithms. Due to the fact that their characteristics are known, it is possible to make an expert evaluation of the results of recommendations resulting from the use of TOPSIS and VIKOR methods.

The Python language and scikit-learn package were used to implement clustering algorithms. In this context, the research assumed default clustering settings, except for the number of clusters k , which is a parameter of the grouping options analyzed. The analysis considered

Table 1
General characteristics of selected clustering algorithms [43,44].

Algorithm (Class)	Description	Advantages	Disadvantages
Agglomerative (Hierarchical)	Starts with each data point as an individual cluster and merges them based on distance.	Simple to understand, can handle different shapes and sizes of clusters.	Sensitive to noise and outliers, computationally expensive for large datasets, not suitable for high-dimensional data.
K-means (Partitioning)	Aims to partition the data into K clusters, where each data point belongs to the cluster with the closest mean value.	Works well with large data sets, is simple and computationally efficient.	Assumes clusters are spherical and of similar size, sensitive to initial cluster centers.
BIRCH (Hierarchical)	Builds a tree-like structure to represent the data, then clusters the leaf nodes.	Handles noisy data well, memory efficient, can deal with outliers, good enough for large datasets.	High dimensional, parameter sensitive data may not be processed well.
Genetic (Evolutionary)	Simulates the process of natural selection to evolve cluster assignments.	Capable of dealing with complex and non-linear data structures and relationships in the data.	May converge to suboptimal solutions, is computationally expensive and sensitive to parameter settings.
K-medians (Partitioning)	Similar to K-means, but uses the median instead of the mean to represent the center of a cluster.	Works well with non-spherical clusters and data with high-dimensional noise, less sensitive to outliers than K-means.	Sensitive to initial cluster centers, more computationally intensive and slower than K-means.

id	device	user	url	type	group	category	action	name	time	initKey
2139592	87144670763015	06c02326-155b-470b-b07d-ac3510a6	https://www.igoras.com.pl/glos-pisn-kreks-416237mrm25-43s.html	event	230401_230612_1686725101_3	ProductPage	selectSize	152	16.06.2023	event-ProductPage-selectSize-152-cenu-gluc
1533978	4164572044873b	0a6bde5-4ed4-410b-a037-054518107460	https://www.igoras.com.pl/au-velosort-ar-geta-litewell-416237mrm156-20s.h	product	default				16.06.2023	product-vitess-velosort
2596020	80445000222552	8a60655-24d0-42d1-963e-46b616174919	https://www.igoras.com.pl/rdosana/sievietes.html	listing	230401_230612_1686725101_10				16.06.2023	listing-igardosana-sievietes
4030903	1156b6c7e764ab	74eb0475-4b10-4155-8453-30d752203b	https://www.igoras.com.pl/sporta-dzempers-hud-3-416237mrm574-4s.h	event	default				16.06.2023	event-ProductPage-thumbClick-cenu-sporta
1897001	4465710506a79a5	30375545-6489-40b3-8ed4-c9a464e132	https://www.igoras.com.pl/rdosana/sievietes.html	listing	default				16.06.2023	accounts-customers-account
1432335	08051962134431b	0a065754-0585-43d1-ac79-79266664a18e	https://www.igoras.com.pl/sporta-dzempers-hud-3-416237mrm574-4s.h	listing	default				16.06.2023	listing-sievietes-ardosana-sievietes-ardosana
2695708	6556262564827a	8f0e2e68-8416-4458-8251-32610718f65	https://www.igoras.com.pl/rdosana/sievietes.html	listing	default				16.06.2023	listing-igardosana-sievietes-ardosana
1423110	3023611eac22071	47b6cda0-8803-47be-ae01-4025902098a	https://www.igoras.com.pl/sporta-dzempers-hud-3-416237mrm574-4s.h	listing	default				16.06.2023	event-sporta-dzempers-hud-3-416237mrm574-4s.h
1488290	875ac8d0c0c0d0	811ab7e7-656c-466b-bada-c5a07d5c0e08	https://www.igoras.com.pl/rdosana/sievietes.html	listing	default				16.06.2023	checkout-checkout-login
3761432	11414508130-0956	6ac36676-ed8a-4984-9892-cb1649190517	https://www.igoras.com.pl/bestelleri-meltones.html	listing	230401_230612_1686725101_10				16.06.2023	listing-bestelleri-meltones
3482982	7031060a0b7201	cc731022-6964-4ed0-b1e4-c4006020249	https://www.igoras.com.pl/sporta-dzempers-hud-3-416237mrm574-4s.h	listing	230401_230612_1686725101_10				16.06.2023	listing-sporta-dzempers-hud-3-416237mrm574-4s.h
4716787	3604d03d01a1e0c0	d971753b-1e6d-496b-bb3c-c5734ec796	https://www.igoras.com.pl/sporta-dzempers-hud-3-416237mrm574-4s.h	listing	lottery_default_from00002023				16.06.2023	listing-sporta-dzempers-hud-3-416237mrm574-4s.h
1572003	34d10181a3984	a088ab-26b6-4951-b17c-84c0baf0ec	https://www.igoras.com.pl/sporta-dzempers-hud-3-416237mrm574-4s.h	product	default				16.06.2023	product-vitess-igoras
1687751	a7ab863b0b11e0b	72a2c1e8-4b3a-47b6-a7b6-80745b473b	https://www.igoras.com.pl/sporta-dzempers-hud-3-416237mrm574-4s.h	product	default				16.06.2023	product-vitess-igoras
1023440	4a024703000071	118a20a-507b-40e6-b471-cd173d9a2082	https://www.igoras.com.pl/sporta-dzempers-hud-3-416237mrm574-4s.h	product	230401_230612_1686725101_4				16.06.2023	product-meltones-igoras
1717287	0a20b0c0a080a00	83d4d4e-5261-45ea-99ac-a1a1c03af0da	https://www.igoras.com.pl/sporta-dzempers-hud-3-416237mrm574-4s.h	product	default				16.06.2023	checkout-checkout-onpage-success
3751666	5d04d4c0c0c0c0	451a66a-703a-4185-b07a-a030f0a0a0	https://www.igoras.com.pl/sporta-dzempers-hud-3-416237mrm574-4s.h	product	230401_230612_1686725101_10	ProductPage	selectColor	Zala	16.06.2023	event-ProductPage-selectColor-Zala-vitess-stept
1101015	51a4d02749584	07a0152-4244-47b6-b5ec-a440f0e091	https://www.igoras.com.pl/sporta-dzempers-hud-3-416237mrm574-4s.h	product	default				16.06.2023	event-bastelleri-vitess-produkty_kategorie-sporta
1572226	499c1414507d5a	5407a20-5620-4e4d-a88a-4f8a6605d49	https://www.igoras.com.pl/sporta-dzempers-hud-3-416237mrm574-4s.h	landing	default				16.06.2023	landing-landing-4-atains
1610516	9ab0a7002270071	874b10a4-9b0d-43aa-a113-c9f0d0b0c5e	https://www.igoras.com.pl/sporta-dzempers-hud-3-416237mrm574-4s.h	checkout	default				16.06.2023	checkout-checkout-cart
3013499	10300915123207	34091652-84e4-401b-b087-70a0d43a4c79	https://www.igoras.com.pl/sporta-dzempers-hud-3-416237mrm574-4s.h	listing	230401_230612_1686725101_10				16.06.2023	listing-bastelleri-sievietes
4479413	03a30b4b-830b0a2	35d1533c-d7f1-4b8b-8381-03a375411704	https://www.igoras.com.pl/sporta-dzempers-hud-3-416237mrm574-4s.h	checkout	lottery_custom_from00002023				16.06.2023	checkout-checkout
1610675	4afa70b00061707	378b0d4d-1a02-445d-9483-7b0a839b3d8	https://www.igoras.com.pl/sporta-dzempers-hud-3-416237mrm574-4s.h	product	default				16.06.2023	product-sievietes-sporta
1550432	734776c0204a406	9a601111-4229-4129-8940-4a090373665	https://www.igoras.com.pl/sporta-dzempers-hud-3-416237mrm574-4s.h	product	default				16.06.2023	account-customer-account-login-referer
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1447183	878300309a7300	81b0803-1833-481b-9331-2446b-c0e790	https://www.igoras.com.pl/sporta-dzempers-hud-3-416237mrm574-4s.h	listing	default				16.06.2023	listing-sievietes-ardosana-sievietes-page-2
1461450	4073083243a0c05b	5a5c5480-09c7-471a-b298-1301125a1036	https://www.igoras.com.pl/sporta-dzempers-hud-3-416237mrm574-4s.h	listing	default				16.06.2023	listing-igardosana-sievietes-page-21
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1466615	1666a0c0a010333	4a5e7003-740d-4122-b7e6-b0e0ba7011	https://www.igoras.com.pl/sporta-dzempers-hud-3-416237mrm574-4s.h	product	default				16.06.2023	product-sievietes-t
1571170	35142b02148a2c63	3a30981b-df5d-4c4d-8030-1a0113050056	https://www.igoras.com.pl/sporta-dzempers-hud-3-416237mrm574-4s.h	listing	default				16.06.2023	listing-sievietes-ardosana-sievietes-page-3
1463218	13046b0a301062	8a3a1e15-070d-4d4f-a0a5-770e0d-c0a054	https://www.igoras.com.pl/sporta-dzempers-hud-3-416237mrm574-4s.h	product	lottery_default_from00002023				16.06.2023	event-ProductPage-selectSize-1-vitess-1
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3424011	14970225-c00704	92a4d551-5361-4c35-9710-3070c4c0201	https://www.igoras.com.pl/sporta-dzempers-hud-3-416237mrm574-4s.h	checkout	230401_230612_1686725101_2				16.06.2023	checkout-checkout-cart
3047992	0a6c2a0c098010	19444652-119-438a-bc15-80937070a028	https://www.igoras.com.pl/sporta-dzempers-hud-3-416237mrm574-4s.h	event	230401_230612_1686725101_10	ProductPage	selectColor	Violeta	16.06.2023	event-ProductPage-selectColor-Violeta-meltones-pi
2723154	15545480b131a0b0	eeb34385-0531-408b-b111-cd153022079a	https://www.igoras.com.pl/sporta-dzempers-hud-3-416237mrm574-4s.h	listing	default				16.06.2023	listing-igardosana-sievietes-accounts-ardosana-ig
2601126	727324767c0871	3a654e14-b118-4d3a-aaf1-8806a176a087	https://www.igoras.com.pl/sporta-dzempers-hud-3-416237mrm574-4s.h	listing	230401_230612_1686725101_10				16.06.2023	listing-sievietes-ardosana-sievietes-ardosana-smiks
1502013	1eac2a0c098010	2402401-3a08-415a-b382-0e01161836	https://www.igoras.com.pl/sporta-dzempers-hud-3-416237mrm574-4s.h	listing	default				16.06.2023	listing-igardosana-sievietes-page-3

Unique IDs of the device. Unique IDs of the user. Full URL of the event occurrence. Detailed information about the event resulting from user activity. Aggregated event information for reporting on user behavior.

Fig. 1. An excerpt from the raw behavioral data.

5 clustering algorithms, calculating clusters for $3 \leq k \leq 10$, for a total of $j = 40$ decision alternatives. Clustering outcomes include several datasets containing:

- assignment of customers to clusters,
- cluster visualizations (UMAP, t-SNE, PCA),
- sizes of clusters,
- values of context-free quality indicators,
- statistics (mean, standard deviation) on customer behavior in clusters.

The raw dataset and clustering reports have been published to the repository [48].

The recommendations and selection of the best clustering alternatives were based on a set of decision criteria, as well as univariate and multivariate analyses described in the following section.

3. Theory

3.1. Decision criteria

Quality metrics are decision criteria in the comparative analysis of clustering methods divided into four characteristics:

- Context-free (CF) includes typical measures of clustering quality commonly used to compare different clustering methods.
- Resource usage (RU) comprises server resource consumption and clustering time.
- Cluster spread (CS) is related to the business requirement that expects clusters to be similar in size.
- Cluster distinguishability (DG) is related to the business requirement of easily distinguishing clusters based on quantitatively described characteristics.

Table 2
Quality metrics (decision criteria) — non-business aspects.

Metric	Characteristics	Description
Silhouette Score [49] (<i>SS</i>)	Context-free (CF)	Quantifies how similar a data point is to its own cluster compared to other clusters, ranging from -1 to 1 , with a high value indicating that the object is well matched to its own cluster and poorly matched to neighboring clusters.
Calinski–Harabasz Index [50] (<i>CHI</i>)	Context-free (CF)	A quantitative measure of cluster separation that compares the dispersion between clusters to the dispersion within clusters.
Davies–Bouldin Index [51] (<i>DBI</i>)	Context-free (CF)	Measures the compactness and separation of clusters, quantifying how well clusters are spread out and how well they are separated from each other.
Clustering time (<i>CT</i>)	Resource usage (RU)	Total clustering time, measured in seconds; the lower the value, the better (clusters are created faster, with shorter resource usage).
Delta CPU_{max} ($dCPU_{max}$)	Resource usage (RU)	The difference between the percentage of maximum CPU usage during clustering minus the percentage of CPU usage of the unloaded server.
Delta MEM_{max} ($dMEM_{max}$)	Resource usage (RU)	The difference between the percentage of maximum memory (MEM) usage during clustering minus the percentage of MEM usage of the unloaded server.

Table 3
Quality metrics (decision criteria) — business aspects.

Metric	Characteristics	Description
Cluster Spread Index (<i>CSI</i>)	Cluster spread (CS)	The difference between the number of customers in the largest cluster minus the number of customers in the smallest cluster, divided by the number of customers in the largest cluster.
Size of the smallest cluster (<i>Smin</i>)	Cluster spread (CS)	The number of clients in the smallest cluster divided by the number of all clustered users.
Standard deviation of cluster sizes (<i>SDCS</i>)	Cluster spread (CS)	Calculated standard deviation of the cluster sizes.
Distinguishability of actions (<i>DA</i>)	Distinguish-ability (DG)	The minimum difference between the average number of actions in neighboring clusters, assuming clusters are ranked by number of actions.
Distinguishability of events (<i>DE</i>)	Distinguish-ability (DG)	The minimum difference between the average number of events in neighboring clusters, assuming clusters are ranked by number of events.
Distinguishability of revenue (<i>DR</i>)	Distinguish-ability (DG)	The minimum difference between the average number of revenue in neighboring clusters, assuming clusters are ranked by number of revenue.

Details of the metrics are shown in Tables 2 and 3 concerning non-business and business aspects.

The study made assumptions about the weights of each decision criterion for the purposes of multi-factor analysis, and it was decided that they were all equal. However, many MCDA methods allow different weights for criteria, further increasing the flexibility of such an approach. The challenge of how weights affect results and rankings can be the subject of further research.

3.2. Single-factor analysis

Most of the analyzed measures aimed at value maximization to increase the quality of clustering. Therefore, to standardize the interpretation of the single-factor analysis, the metrics aimed at minimizing the value were modified through inversion. An example is the Davies–Bouldin Index, where Inverted Davies–Bouldin Index (*IDBI*) was introduced and calculated according to formula:

$$IDBI = \frac{1}{DBI} \quad (1)$$

where: *DBI* is the computed value of the Davies–Bouldin Index.

The values of the metrics used were standardized for visualization and comparability of the charts during the single-factor analysis:

$$sX_i = \frac{X_i}{X_{max}} * 100 \quad (2)$$

where: sX_i is the standardized value of the metric X_i , X_{max} is the highest value of the metric X among all (i)-clustering options.

The single-criteria analysis focused on calculating the rankings of the clustering methods, considering only a single factor. Therefore, it was possible to compare the results and verify the sensitivity of the recommendation of the best solution depending on the choice of the analyzed feature of the studied algorithms.

3.3. Multi-factor analysis

The purpose of evaluating the clustering algorithms using MCDA was to identify the optimal method adopted to cluster e-commerce customers based on data on their behavior. The VIKOR and TOPSIS methods were selected for analysis to confront the results obtained and draw conclusions.

3.3.1. VIKOR

The goal of VIKOR is to find a compromise solution that is as close as possible to the ideal option while taking into account the distance to the worst solution. This method facilitates the selection of a compromise option to balance multiple criteria and reflect the preferences and priorities of the decision-maker. The method is designed to solve problems stated as [52]:

$$mco\{(f_{ij}(A_j), j = 1, \dots, J), i = 1, \dots, n\} \quad (3)$$

where: A_j stands for j th alternative, f_{ij} is the value of the i th decision factor function for A_j ; J is the number of available options, n is the number of criteria; mco stands for multi-criteria decision procedure operator.

The application of the method involves the following activities:

1. Determination of the extreme values (best - f_i^* and worst - f_i^-) of f_{ij} for each i : $f_i^* = \max_j f_{ij}$, $f_i^- = \min_j f_{ij}$, when i th factor is expected to be maximized, $f_i^* = \min_j f_{ij}$, $f_i^- = \max_j f_{ij}$, when i th factor is expected to be minimized;
2. Determination of weights (w_i) for the following i decision criteria;
3. Calculation of the S_j value that represents the closeness to the ideal:

$$S_j = \sum_{i=1}^n w_i \frac{(f_i^* - f_{ij})}{(f_i^* - f_i^-)}; \quad (4)$$

4. Calculation of the R_j value that represents the closeness to the worst option:

$$R_j = \max_i [w_i \frac{(f_i^- - f_{ij})}{(f_i^- - f_i^*)}]; \quad (5)$$

5. Calculation of the value of Q_j that represents Utility Measure:

$$Q_j = v \frac{(S_j - S^*)}{(S^- - S^*)} + (1 - v) \frac{(R_j - R^*)}{(R^- - R^*)}, \quad (6)$$

where: $S^* = \min_j S_j$, $S^- = \max_j S_j$, $R^* = \min_j R_j$, $R^- = \max_j R_j$, v is a weight for the strategy of maximum group utility, and $(1 - v)$ is the weight of the individual regret;

6. Sorting of the alternatives, according to the Q -value in ascending order to obtain a ranking of alternatives.

This research omits the next steps of the VIKOR method since the aim was to develop a ranking of alternatives, leading to the proposal of a compromise solution. The full application of the method will be the subject of future research.

3.3.2. TOPSIS

TOPSIS is based on the concept that the preferred alternative should be closest to the positive ideal solution while simultaneously being farthest from the negative. The method helps decision-makers to systematically evaluate and rank alternatives by using a variety of criteria. The scheme for TOPSIS method is as follows [53]:

1. Determination of the decision matrix $F = \{f_{ij}\}$;
2. Calculation of the normalized (e.g. using the normalization with the Euclidean norm [54]) decision matrix $R = \{r_{ij}\}$:

$$r_{ij} = \frac{f_{ij}}{\sqrt{\sum_i f_{ij}^2}} \quad (7)$$

3. Calculation of the weighted decision matrix $V = \{v_{ij}\}$:

$$v_{ij} = w_i * r_{ij} \quad (8)$$

4. Calculation of the A^* value that represents the ideal solution and A^- that represents the negative-ideal solution:

$$A_i^* = \min_j (v_{ij}) \quad A_i^- = \max_j (v_{ij}) \quad (9)$$

5. Calculation of the separation measures S_j^* and S_j^- for each alternative j :

$$S_j^* = \sqrt{\sum_i (v_{ij} - A_i^*)^2} \quad S_j^- = \sqrt{\sum_i (v_{ij} - A_i^-)^2} \quad (10)$$

6. Calculation of the relative closeness C_j to the ideal solution for each alternative:

$$C_j = \frac{S_j^-}{(S_j^- + S_j^*)} \quad (11)$$

7. Sorting of the alternatives, according to the C -value in descending order to obtain a ranking of alternatives.

In the final stage of the study, a sensitivity analysis of the rankings of alternatives obtained by both MCDA methods was performed to verify the impact of the selection of a set of features (decision criteria) on the recommendation of the clustering option. For possible combinations of quality characteristics, rankings of alternatives were calculated using VIKOR and TOPSIS methods. In this context, 15 rankings were obtained for each decision-making method, and the results allowed for discussion and conclusions.

The presented framework has been designed to select and evaluate clustering methods for the behavior-based grouping of e-commerce customers. It is used to determine the optimal parameterization of clustering algorithms and assist in the selection. The method is validated using a combination of known quality factors, supplemented by additional measures derived from business requirements. The results obtained show that the solution can optimize the decision-making process and increase the quality of segmentation for e-commerce purposes. Insights into the properties of clustering methods create future opportunities for further research using different types of algorithms and applied methods.

4. Results

The research is divided into three sections, covering the clustering of behavioral data and a two-step analysis of both selected MCDMs. Subsequent sections describe the results, single decision factors, and multi-criteria analysis, respectively.

4.1. Clusterization

The grouping was based on behavioral data of the online customers collected over 2 months using AIM^2 platform. The dataset has been anonymized and meets all user privacy requirements.

At this stage of the research, 210,274 user sessions were identified. The raw data set was preprocessed to be used for customer segmentation. Next, five clustering methods were selected and performed with different values of the parameters to determine the number of clusters (k). The research was conducted for $3 \leq k \leq 10$ due to the business specificity of using generated clusters. This decision was based on the requirements for the minimum number of customers in a cluster, increasing the rationality of designing a UI variant.

The visualization of clusters (for $k = 6$) using the t-SNE approach [55] is shown in Fig. 2. The distribution of clusters based on visualization is interpreted in general terms since the method only represents high-dimensional data in a low-dimensional space. At this level, the atypical distribution of clusters for the *Genetic* method can be observed. In this case, there are no homogeneous user groups to suggest clustering useful from a business perspective. However, there are some similarities in the visualization of clusters obtained by the other methods. Even though the labels of the clusters are different, the common features are visible. This is evident in the clusters filling the top left of the visualization and centered at the bottom of the graphs. The preliminary analysis leads to the conclusion that the results

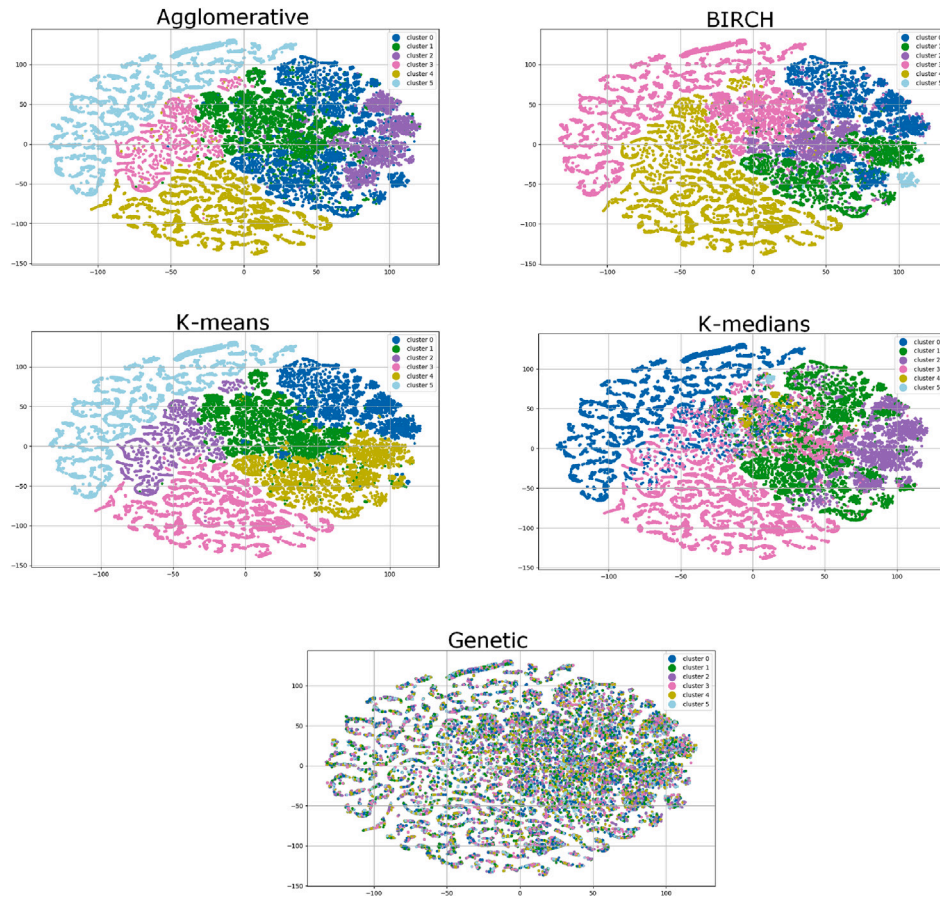


Fig. 2. Visualization of the clusters (t-SNE, $k = 6$).

Table 4
Number of customers in clusters ($k = 6$).

Method label	Agglomerative	BIRCH	K-means	K-medians	Genetic
cluster 0	13,228	8424	8333	15,593	9,976
cluster 1	10,454	8,485	9,847	14,093	10,072
cluster 2	5,806	5,330	7,117	8,776	10,342
cluster 3	7,405	18,025	12,417	20,665	10,088
cluster 4	12,131	19,667	11,307	1,029	10,069
cluster 5	11,594	687	11,597	462	10,071

obtained for the *Genetic* method differ significantly from the indicators calculated for other algorithms.

In addition to the visualizations, it is also worth noting the number of e-commerce customers assigned to each cluster, depending on the grouping method. This is key information from a business perspective. Personalizing communications with customers requires consideration of the cost of such activities. For this reason, it is worth aiming for clusters that are as comparable in size as possible, so that dedicated content has a chance of reaching a similar number of recipients. Examples of the results (for $k = 6$) obtained are presented in Table 4. A total of 60,618 unique e-commerce customers were clustered. It is worth noting the different distribution of cluster sizes depending on the method used. With *Genetic*, the clusters are practically equally numerous, while *K-medians* has generated both the most numerous and the least numerous clusters among all analyzed algorithms. This is the first confirmation of the conclusions drawn from the cluster visualization. This feature, which differentiates clustering methods, has been analyzed in detail because this feature affects the values of some metrics.

4.2. Single-factor analysis

4.2.1. Context-free quality

In the research described in the literature, the choice of the optimal method is based on a single characteristic related to clustering effects. Classical quality indicators such as SS, DBI, CHI, and DI, are particularly popular. The indicators are applied regardless of the business context and the characteristics of the datasets. The values of the metrics depend on clustering parameterization, particularly the value of k .

During the described research, the values of SS, DBI, and CHI indices were calculated. Due to differences in interpretation (when optimizing clustering, the goal is to maximize the values of the SS and CHI and minimize the value of the DBI index), the Inverted DBI Index (IDBI) was used. The different values obtained for the presentation of the results were standardized in the range of $< 0, 100 >$, and the analysis of the graphs allows the evaluation of clustering methods based on single factors.

In the case of SS index (Fig. 3, graph A), the best values are obtained after using *Agglomerative* and *K-means*. The lowest standardized values of the metric are obtained by the *Genetic* clustering method but do not fall below 70, so they are still at a relatively high level.

The case is different when analyzing CHI and IDBI indicators as reported in Fig. 3, graphs B and C. The values of the indicators for the *Genetic* method are slightly above zero, showing the unacceptable clustering results. The relationships between the other techniques are not significantly different, although the CHI metric shows an advantage of *K-medians* over *BIRCH*, while the IDBI metric (for $k \geq 6$) indicates the opposite recommendation. Therefore, if any of these measures were considered in the process of selecting a clustering method, the *Genetic* algorithm option would have to be rejected.

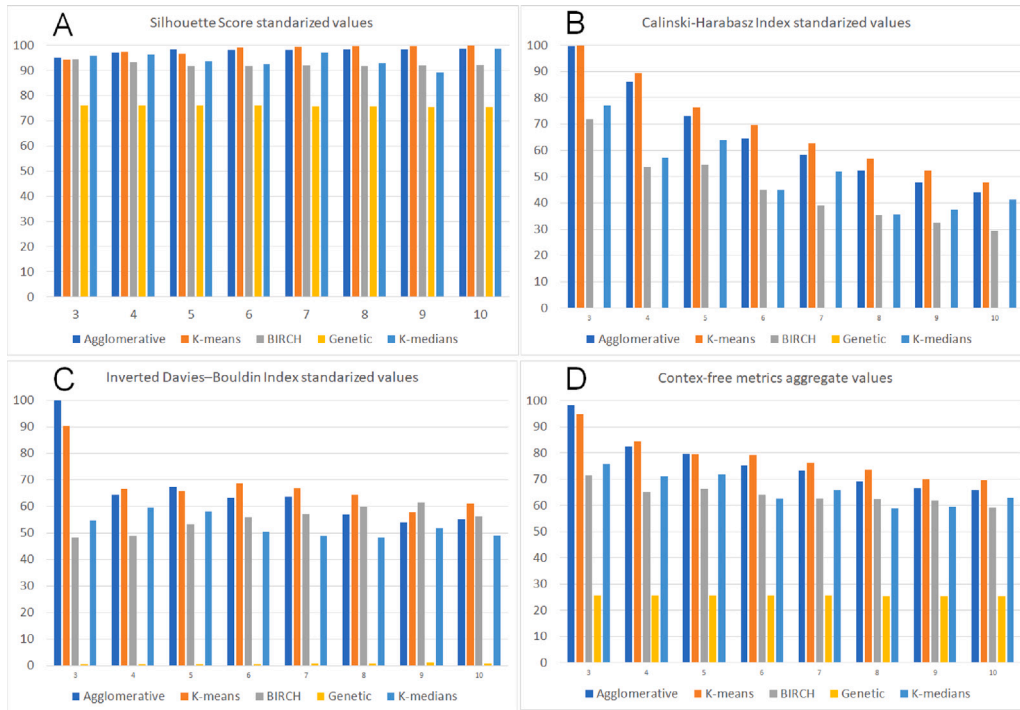


Fig. 3. Comparison of values of context-free (CF) clustering quality indicators.

The decision to select clustering method based on a single quality indicator is risky since different conclusions can be drawn depending on the value of the parameter k . The solution is to define a value that combines multiple indicators through standardization and aggregation. After assuming weights for each indicator, an aggregate value is calculated considering different aspects of clustering quality (Fig. 3, graph D). In this context, the *K-means* and *Agglomerative* algorithms are recommended but the final choice is not unambiguous. This method allows a deeper insight into the decision-making process but is still limited to a single quality characteristic.

4.2.2. Computing resources

An analogous analysis was performed for the second clustering characteristic. A total of three metrics were considered and the results are shown in Fig. 4. As before, the evaluation of clustering method is highly dependent on the measure.

The lower computational complexity of the *K-means* and *BIRCH* methods (Fig. 4, graph A) leads to the best results considering clustering time. *Agglomerative* clustering takes the longest time to compute, as justified by the design of the algorithm.

The case is different for the metric describing the increase in CPU resource consumption during clustering ($dCPU_{max}$). Fig. 4, graph B shows that the differences are relatively small, and the best method in this aspect (*BIRCH*) is not much better than the worst one (*K-medians*). Therefore, the $dCPU_{max}$ index is not a feature strongly differentiating clustering method since the usefulness in the decision-making process is limited. This conclusion should lead to a modification of decision factors in future research.

Interesting results were obtained using the incremental memory usage index (Fig. 4, graph C). *Genetic* and *Agglomerative* were the best and worst clustering, respectively. This was contrasted using the results of the indicators previously analyzed, where *Genetic* was rated strongly negative. The situation shows that the selection of a single decision criterion without considering other factors leads to highly biased conclusions.

The aggregated evaluation of the methods (Fig. 4, graph D) determines the best solution in terms of computing resources but the

results are inconclusive. While *Agglomerative* clustering usually has the lowest score and can be described as the worst solution, it is difficult to identify the best method because it varies depending on the value of the k parameter. This situation confirms the dependence of the values of the indicators describing the quality of clustering on the algorithm and the initial parameters.

4.2.3. The spread of clusters

From a business perspective, clusters grouping e-commerce customers should be of a similar size. However, the research and analysis of cluster spread metrics showed that the requirement should be subject to additional constraints. For indicators based on the standard deviation of cluster size (sSD) and spread (sCSI), the *Genetic* method performed the best compared to others (Fig. 5, graph A).

The dominance of the method was clear for the size of the smallest cluster but to a lesser extent than other metrics (Fig. 5, graph B).

The aggregation of indicators related to cluster spread (Fig. 5, graph C) is not surprising, in the context of the previously described results. *Genetic* seems to be the best solution from the point of view of business requirements, but such a conclusion contradicts the results of the analysis of the previous quality features. Since previously this method was an outlier, the analysis was repeated by omitting it from the calculations (Fig. 5, graph D). Under this new assumption the best results are obtained by the *K-means* algorithm but for some values of the parameter k , the *Agglomerative* method is better, confirming the conclusions of the other group of quality features.

4.2.4. Distinguishability of clusters

The analysis of the ability to distinguish clusters was based on three metrics related to the average values of user characteristics: number of actions, events, and revenue. The calculation is carried out after clustering and is based on reports containing key information on the customer segments generated. A visualization of the standardized values of the metrics in this characteristic is shown in Fig. 6.

It can be observed that the distinguishability of clusters decreases with an increasing value of the k parameter. Therefore, the number of variants should be considered with caution when planning to use

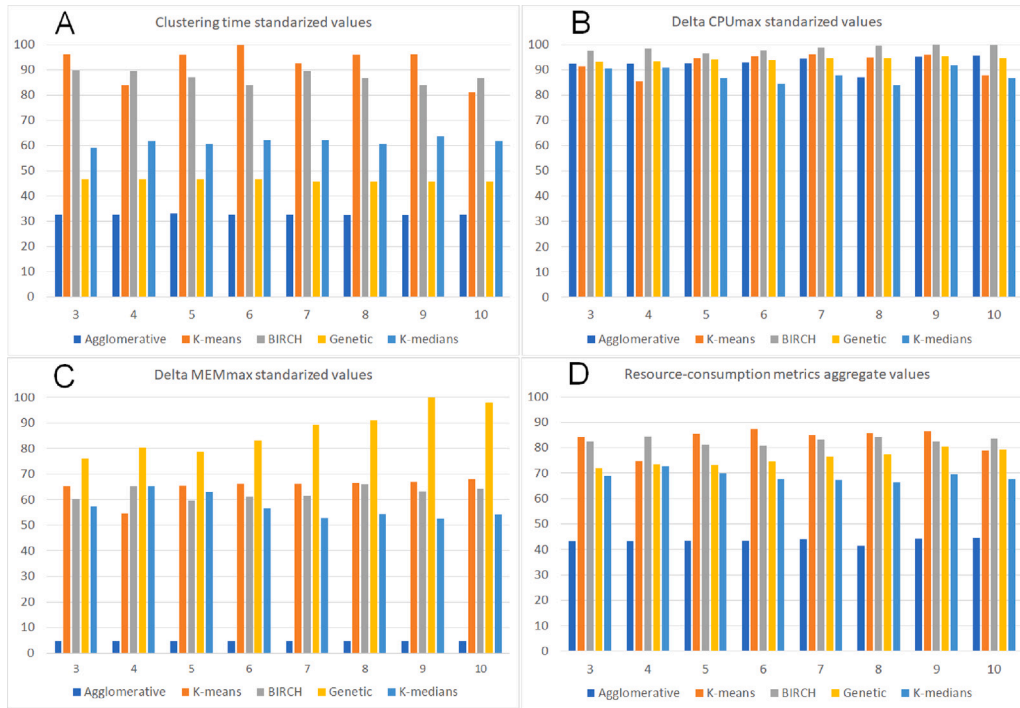


Fig. 4. Comparison of values of computing resources usage (RU) indicators.

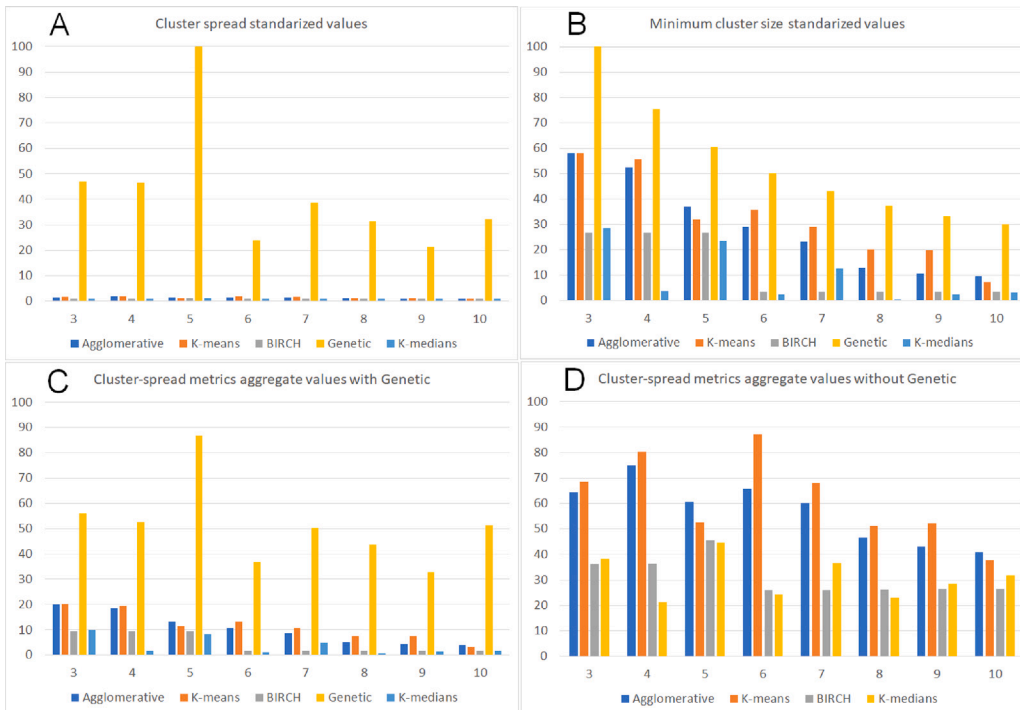


Fig. 5. Comparison of values of clusters' spread (CS) indicators.

the results to serve UI variants. Increasing the number of variants may lead to better personalization and customization but problems arise due to the high similarity of user behavior of clusters with low distinguishability.

In analyzing the results, *BIRCH* and *K-medians* clustering provide high distinguishability at $k = 3$, while *K-means* does so at $k = 4$. Regardless of the number of clusters, the distinguishability of the clusters obtained by the *Genetic* method is practically zero and the conclusion is supported by aggregate values (Fig. 6, graph D).

The described set of metrics can be freely extended by differentiating clusters based on new features. This gives an opportunity to analyze the business needs arising from e-commerce segmentation. This group of measures provides valuable guidance to decision-makers by verifying the justification for increasing the number of segments. It is characterized by a large variation of values depending on the parameter k . Therefore, this approach can be useful when showing the optimal number of clusters from a narrow range of values. In addition, results revealed a very significant disadvantage of the *Genetic* method, which

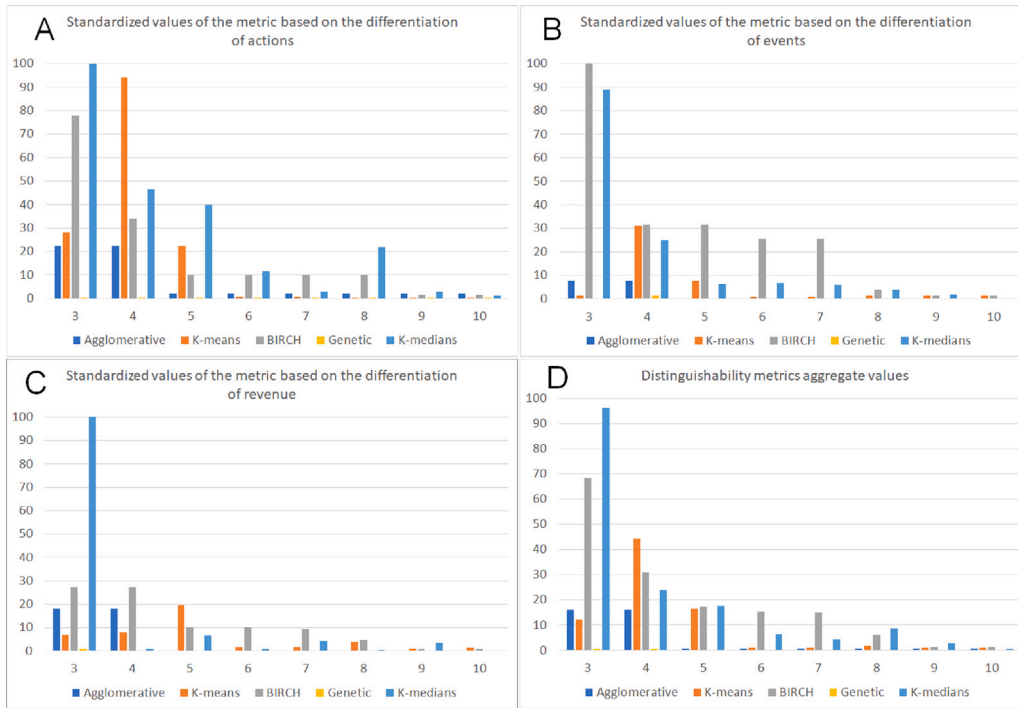


Fig. 6. Comparison of values of clusters' distinguishability (DG) indicators.

was the best according to the cluster spread metrics. This means that in order to decide which clustering algorithm to choose, it is necessary to rely on several metrics that reflect different quality characteristics. Therefore, it seems reasonable to use MCDA methods to comprehensively evaluate different aspects of clustering and make an optimal choice. This conclusion provides a practical justification for conducting an analysis using VIKOR and TOPSIS.

4.3. Multi-factor analysis

4.3.1. VIKOR analysis

The first element of the research was to rank the decision alternatives considering the criteria. Calculations were based on VIKOR method, assuming equal weights for decision criteria ($\forall_i w_i = 1$) and strategy of maximum group utility for individual regret ($v = 0.5$). Therefore, the basic equations for calculating key indicator values take simplified form:

- the closeness to the ideal:

$$S_j = \sum_{i=1}^n \frac{(f_i^* - f_{ij})}{(f_i^* - f_i^-)}; \quad (12)$$

- the closeness to the worst option:

$$R_j = \max_i \left[\frac{(f_i^* - f_{ij})}{(f_i^* - f_i^-)} \right]; \quad (13)$$

- utility measure:

$$Q_j = 0.5 \frac{(S_j - S^*)}{(S^- - S^*)} + 0.5 \frac{(R_j - R^*)}{(R^- - R^*)}; \quad (14)$$

The final results of the calculations are shown in Table 5. The best score for clustering alternative uses the *K-medians* method with $k = 3$ and the most common algorithm in the TOP10 ranking is *BIRCH*.

The rankings may change when the assumptions regarding the decision criteria are varied. Therefore, additional sensitivity analysis of the result can be performed depending on changes in the set of clustering quality metrics. The research was conducted by ranking the alternatives after applying different combinations of quality characteristics (Table 6):

Table 5

TOP10 decision alternatives according to the VIKOR method ($j=40, i=12$).

Rank	Method	k	S_j	R_j	Q_j
1	K-medians	3	3.461	0.902	0.013
2	BIRCH	5	4.950	0.900	0.170
3	K-means	4	4.150	0.921	0.182
4	K-means	5	4.524	0.922	0.233
5	BIRCH	4	5.029	0.925	0.305
6	K-medians	5	5.566	0.937	0.425
7	BIRCH	7	5.108	0.969	0.533
8	BIRCH	6	5.126	0.969	0.535
9	BIRCH	3	3.812	1.000	0.540
10	BIRCH	8	5.369	0.969	0.563

- CF — Context-Free;
- RU — Resource Usage;
- CS — Cluster Spread;
- DG — Distinguishability;

A total of four attributes provided 15 possibilities, considering 1, 2, 3, or 4 groups of indicators. The results were calculated for all alternatives ($k \geq 3$) and the set restricted to options with $k \geq 4$.

Considering all the decision options, the *K-medians* method with parameter $k = 3$ proved to be the best clustering option in most cases. This method is the best in rankings due to the distinguishability of clusters.

On the other hand, *K-means* clustering is most common (9/15) when combinations with $k = 3$ are omitted. However, the recommendations address different values of k , and none of them is dominant. It should also be noted that the *BIRCH* method is often (5/15) recommended as the best approach. The *Genetic* algorithm achieves the best result when cluster spread metrics are considered, confirming the observations from the single-factor analysis. The results confirm the importance of sensitivity analysis and prove that reducing the number of decision factors has a significant impact on the recommendations for clustering method. This conclusion is relevant for the practical application of MCDM in the design of customer segmentation tools for e-commerce.

Table 6

Best alternatives with different groups of decision criteria — VIKOR. Groups of decision factors are presented in the first four columns.

CF	RU	CS	DG	Decision ($k \geq 3$)	Decision ($k \geq 4$)
X	X	X	X	K-medians (3)	K-means (5)
X	X	X		K-means (6)	K-means (6)
X		X	X	K-medians (3)	K-means (4)
X	X		X	K-medians (3)	BIRCH (4)
	X	X	X	K-medians (3)	K-means (5)
X	X			BIRCH (3)	K-means (6)
X		X		K-means (6)	K-means (6)
X			X	K-medians (3)	BIRCH (4)
	X		X	K-medians (3)	BIRCH (4)
	X	X		K-means (6)	K-means (6)
		X	X	K-medians (3)	K-means (4)
X				K-means (4)	K-means (4)
	X			BIRCH (10)	BIRCH (10)
		X		Genetic (10)	Genetic (10)
			X	K-medians (3)	BIRCH (4)

Table 7

TOP10 decision alternatives according to the TOPSIS method ($j = 40$, $i = 12$).

Rank	Method	k	S_j^+	S_j^-	C_j
1	K-medians	3	0.518	1.338	0.721
2	BIRCH	3	0.847	1.027	0.548
3	K-means	4	0.958	0.926	0.492
4	BIRCH	4	1.013	0.708	0.411
5	K-means	5	1.086	0.721	0.399
6	K-means	3	1.150	0.749	0.394
7	BIRCH	5	1.112	0.690	0.383
8	K-medians	5	1.154	0.693	0.375
9	Agglomerative	4	1.128	0.666	0.371
10	K-means	6	1.253	0.726	0.367

4.3.2. TOPSIS analysis

TOPSIS was applied in a similar method to VIKOR and the decision matrix is simplified as:

$$v_{ij} = r_{ij} \quad (15)$$

In the first step, the ranking of alternatives was calculated considering the decision criteria, and the sensitivity analysis of the decision was performed. *K-medians* clustering with $k = 3$ was considered the best option (Table 6). In this aspect, both MCDA methods reported the same result. However, there are differences when comparing the entire TOP10 ranking. When using TOPSIS *K-means* clustering appeared most often (4 times), and there was a smaller percentage of options using the *BIRCH* algorithm. In addition, one alternative used the *Agglomerative* method. Therefore, TOPSIS generated more diverse recommendations than VIKOR (see Table 8).

The sensitivity analysis of the best alternative is shown in Table 7. The results are similar to VIKOR when the all decision options are considered. After deeper analysis it can be found that the same results were observed only when *K-median* and $k = 3$ were recommended as the best option. This is another argument pointing to the strong influence of distinguishability metrics on the ranking position of the alternative. In other cases, the recommendations differed — even if the clustering algorithm is the same, the value of k varies.

Another interesting observation was revealed by limiting the alternatives to $k \geq 4$. In this context, the dominance of the *K-means* and $k = 4$ was reported as the best in 13 combinations. Twice the *Genetic* algorithm was ranked first, when a large part of the results was contributed by cluster spread metrics. Despite the greater variation in the TOP10 ranking when TOPSIS was used, the method provided a clearer indication when sensitivity analysis was performed. This low sensitivity of recommendations to the decision criteria provided a strong basis for considering the *K-means* with $k = 4$ as the optimal solution for segmenting e-commerce customers in the described

Table 8

Best alternatives with different groups of decision criteria — TOPSIS. Groups of decision factors are presented in the first four columns.

CF	RU	CS	DG	Decision ($k \geq 3$)	Decision ($k \geq 4$)
X	X	X	X	K-medians (3)	K-means (4)
X	X	X		K-means (4)	K-means (4)
X		X	X	K-medians (3)	K-means (4)
X	X		X	K-medians (3)	K-means (4)
	X	X	X	K-medians (3)	K-means (4)
X	X			K-means (3)	K-means (4)
X		X		K-means (4)	K-means (4)
X			X	K-medians (3)	K-means (4)
	X		X	K-medians (3)	K-means (4)
	X	X		Genetic (3)	Genetic (4)
		X	X	K-medians (3)	K-means (4)
X				Agglomerative (3)	K-means (4)
	X			K-means (9)	K-means (9)
		X		Genetic (3)	Genetic (4)
			X	K-medians (3)	K-means (4)

experiment, specifically when groups are to be served dedicated user interface variants.

5. Discussion and conclusions

In conclusion, the biggest challenge in e-commerce segmentation was the volume and variety of data. E-commerce platforms collect vast information about customer demographics, browsing behavior, and purchase history. However, extracting meaningful insights from the data require sophisticated grouping algorithms. In this context, clustering algorithms based on machine learning are a good solution. However, several decision factors usually come into play when evaluating such e-commerce segmentation techniques.

The method proposed in this research divided clustering quality metrics into context-free, resource usage, cluster spread, and cluster distinguishability. However, even with a set of characteristics and quality metrics (decision criteria), it is necessary to face the problem of identifying the optimal method, given specific clustering objectives and applications. This is also a challenge from a business performance perspective for systems based on e-commerce customer segmentation since clustering method significantly affected the efficiency of the solution [56].

The most common method of making decisions based on a single metric did not appear to be optimal, as evidenced by described results. The analysis obtained for 12 different metrics showed that the recommendation of the best clustering option strongly depended on the decision criterion. This was obvious because clustering methods possessed different advantages and disadvantages. Therefore, a decision model based on a single-factor might be unreliable and did not necessarily show the best solution.

Alternatively, methods from the MCDA group can be used, as demonstrated by the VIKOR and TOPSIS approaches. The concept possessed the advantage of flexibility in the selection of decision factors and the possibility of additional parameterization using weights. By using multi-criteria analysis, different features of clustering methods were included in the decision process to increase the chance of identifying the optimal option. However, the recommendations obtained using analyzed methods were different as reported in this research. To increase the reliability of the decision, it is worth considering the analysis of the collected dataset with different methods and comparing the results.

Analyzing the effects of single-criteria and multi-criteria analyses, it can also be observed that in the context of the developed business application aimed at serving multi-variant UIs in e-commerce, clustering with low values of the k parameter must be approached with skepticism. The results showed that the metrics used favored

options with lower k values, as evidenced by the high placement of such alternatives in most rankings with different combinations of decision factors. This was a limitation of the proposed method but could be minimized when the lowest number of outcome clusters was determined by expert judgment.

It is worth noting the data set analyzed in this research. The study used behavioral information obtained from customers of one online store. Users of other stores would have provided different information differently, potentially leading to another recommendation for clustering method. However, the proposed method could be applied to any other e-commerce business application based on customer segmentation. The decision criteria can be modified and adapted to the business needs, since MCDA allows a comprehensive approach to different aspects of clustering quality.

Even though the use of multi-criteria analysis to select the optimal clustering method was not widely used, the conducted research confirmed the effectiveness of the method. However, some limitations were identified and considered. An undeniable challenge was the choice of decision criteria. In the described case, a total of 12 factors were considered, but only a few were significant. Therefore, considering the criteria to keep or discard was important. Another limitation is the need to determine the weights of decision criteria or groups. Sometimes it is a subjective evaluation, but crucial for the final recommendation. One possible solution is to deepen the sensitivity analysis, but even this does not guarantee a fully objective recommendation. An in-depth analysis of the impact of weights on recommendations can be an area for further research.

The research described does not exhaust the potential of the issue of selecting clustering methods for segmentation in e-commerce. Further work will include the evolution of the approach — in terms of the clustering techniques used and their parameters, the multi-criteria analysis methods used, and the datasets. An additional direction for research is to verify the sensitivity of recommendations based on the specific behavior of e-commerce customers.

Considering the importance of clustering users of e-commerce systems, the optimal choice of the method is crucial for business efficiency. The described research showed the weaknesses of decisions based on single-criteria analyses and can encourage greater use of multi-criteria analysis in the evaluation and selection of clustering methods in a variety of practical applications.

CRedit authorship contribution statement

Adam Wasilewski: Writing – original draft, Methodology, Investigation, Conceptualization. **Krzysztof Juszczyszyn:** Validation. **Vera Suryani:** Writing – review & editing.

Declaration of competing interest

The authors declare no competing interests.

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Data availability

Data supporting the results of this study are available for download from the repository: <https://doi.org/10.7910/DVN/MDONDZ> [48].

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